

Лабораторная работа № 9. Использование протокола STP. Агрегирование каналов

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Цель работы

Изучение возможностей протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.

Задание

Задание

1. Сформируйте резервное соединение между коммутаторами msk-donskaya-sw-1 и msk-donskaya-sw-3.
2. Настройте балансировку нагрузки между резервными соединениями.
3. Настройте режим Portfast на тех интерфейсах коммутаторов, к которым подключены серверы.
4. Изучите отказоустойчивость резервного соединения.
5. Сформируйте и настройте агрегированное соединение интерфейсов Fa0/20 – Fa0/23 между коммутаторами msk-donskaya-sw-1 и msk-donskaya-sw-4.
6. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

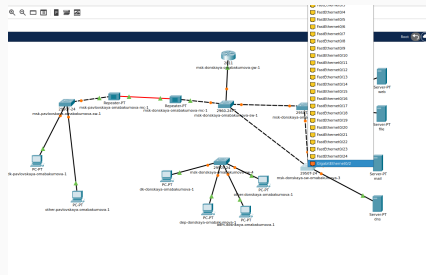


Рис. 1: Замена соединения между коммутаторами

Резервное соединение

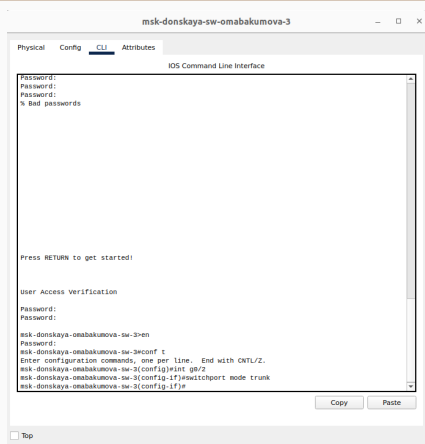


Рис. 2: Порт стал транковым

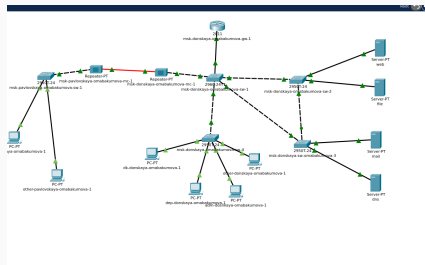


Рис. 3: Соединение между коммутаторами msk-donskaya-omabakumova-sw-1 и msk-donskaya-omabakumova-sw-4

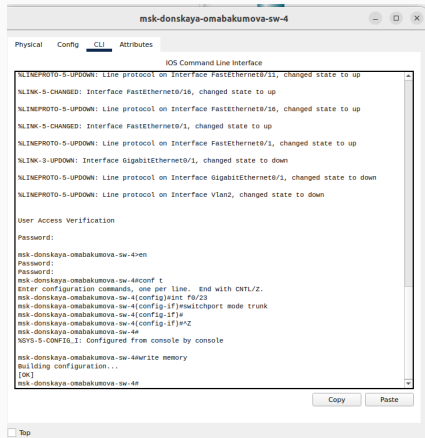


Рис. 4: Активация в транковом режиме

Резервное соединение

```
dk-donskaya-omskumova-1

Physical  Config  Desktop  Programming  Attributes

NetworkManager

root@kali:~/Desktop# ./mail-test.py
C:\>ipconfig

FastEthernet0 Connection (default port)

    connection-specific dns suffix...
    Link-local IPv6 Address . . . . . FE80::20F:FEAF::4E5
    IPv6 Address. . . . . ::
    IPv6 Address. . . . . 10.128.0.31
    Subnet Mask . . . . . 255.255.255.0
    Default Gateway . . . . . 10.128.0.1

Bluetooth Connection:

    connection-specific dns suffix...
    Link-local IPv6 Address . . . . . ::
    IPv6 Address. . . . . ::
    IPv6 Address. . . . . 0.0.0.0
    Subnet Mask . . . . . 0.0.0.0
    Default Gateway . . . . . ::

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.3

Pinging 10.128.0.3 with 32 bytes of data:

Reply from 10.128.0.3: bytes=32 time=1ms TTL=127
Reply from 10.128.0.3: bytes=32 time=1ms TTL=127
Reply from 10.128.0.3: bytes=32 time=1ms TTL=127
Reply from 10.128.0.3: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping mail.donskaya.ru

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
```

Рис. 5: Пингование mail и web

Резервное соединение

```
C:\>ping mail.donskaya.ru

Pinging 10.120.0.4 with 32 bytes of data:

Reply from 10.120.0.4: bytes=32 time=0ms TTL=127
Reply from 10.120.0.4: bytes=32 time=0ms TTL=127
Reply from 10.120.0.4: bytes=32 time=0ms TTL=127
Reply from 10.120.0.4: bytes=32 time=0ms TTL=127

Ping statistics for 10.120.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping www.donskaya.ru

Ping request could not find host www.donskaya.ru. Please check the name and try again.
C:\>ping www.donskaya.ru
Ping request could not find host www.donskaya.ru. Please check the name and try again.
C:\>ping 10.120.0.2

Pinging 10.120.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127

Ping statistics for 10.120.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.120.0.2

Pinging 10.120.0.2 with 32 bytes of data:

Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127

Ping statistics for 10.120.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping www.donskaya.ru

Pinging 10.120.0.2 with 32 bytes of data:

Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127

Ping statistics for 10.120.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 6: Пингование mail и web

Резервное соединение

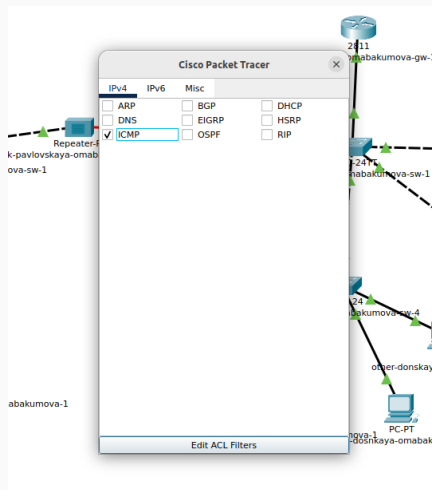


Рис. 7: Фильтрация типа пакета в симуляции

Резервное соединение

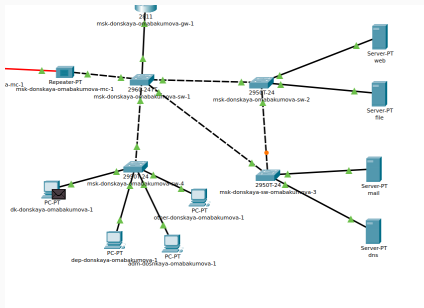


Рис. 8: Пакет от оконечного устройства

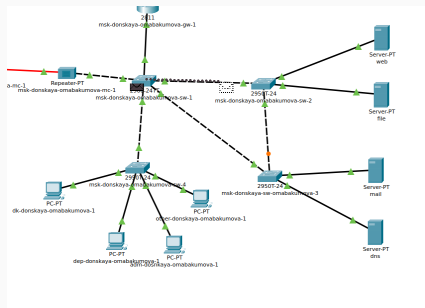
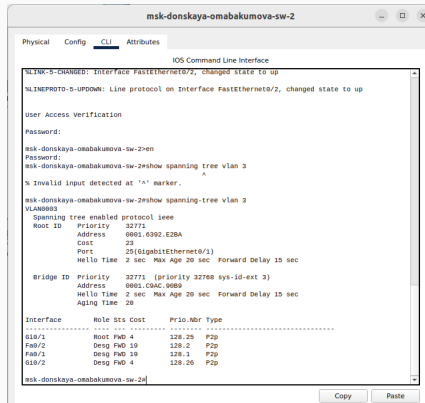


Рис. 9: Успешный проход через msk-donskaya-omabakumova-sw-2

Резервное соединение



The screenshot shows a terminal window titled "msk-donskaya-omabakumova-sw-2" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and their results:

```
msk-donskaya-omabakumova-sw-2#show spanning-tree vlan 3
Spanning tree enabled protocol ieee
Root ID    Priority    32771
           Address    0001.0392.E2BA
           Cost        20
           Port        25(GigabitEthernet0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

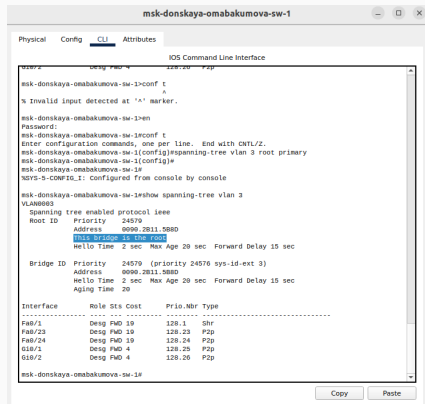
Bridge ID   Priority    32771 (priority 32768 sys-id-ext 3)
           Address    0001.C9AC.9080
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20
```

Interface	Role	Sts	Cost	Prio	Nbr	Type
Gi0/1	Root	PwD	4	128.25		P2p
Fa0/2	Desg	PwD	19	128.2		P2p
Fa0/1	Desg	PwD	19	128.1		P2p
Gi0/2	Desg	PwD	4	128.26		P2p

msk-donskaya-omabakumova-sw-2#

Рис. 10: Состояние протокола STP для vlan 3

Резервное соединение



The screenshot shows a network configuration window titled "msk-donskaya-omabakumova-sw-1". The "CLI" tab is selected, displaying the "IOS Command Line Interface". The terminal output shows the following commands and results:

```
msk-donskaya-omabakumova-sw-1>conf t
% Invalid input detected at '^' marker.
msk-donskaya-omabakumova-sw-1>en
Password:
msk-donskaya-omabakumova-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-omabakumova-sw-1(config)#
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-1#show spanning-tree vlan 3
VLAN0003
Spanning tree enabled protocol ieee
Root ID    Priority    24579
Address    0000.2811.588D
This Bridge is the root
Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

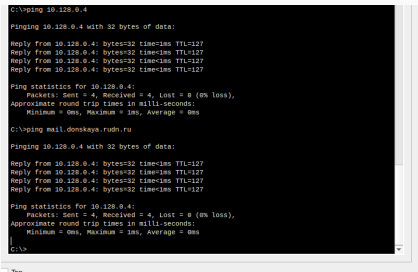
Bridge ID  Priority    24579 (priority 24576 sys-id-ext 3)
Address    0000.2811.588D
Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time 20

Interface  Role Sts Cost      Prio.Nbr Type
-----
Fa0/1      Desg FWD 19      128.1  Shr
Fa0/23     Desg FWD 19      128.23 P2p
Fa0/24     Desg FWD 19      128.24 P2p
Gi0/1      Desg FWD 4      128.25 P2p
Gi0/2      Desg FWD 4      128.26 P2p

msk-donskaya-omabakumova-sw-1#
```

At the bottom of the window, there are "Copy" and "Paste" buttons.

Рис. 11: Настройка в качестве корневого коммутатора



```
C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 12: Пингует для устранения ошибок

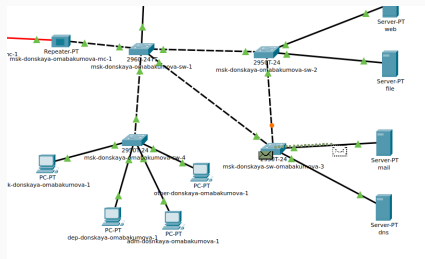


Рис. 13: пакет успешно прошел по нужному маршруту до mail

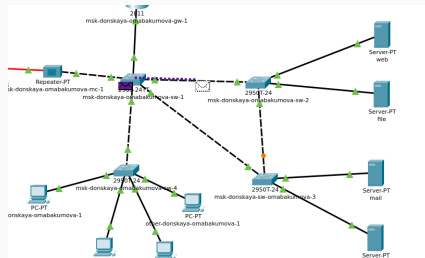


Рис. 14: Пакет успешно прошел по нужному маршруту до web

```
msk-donskaya-omabakumova-sw-2>en
Password:
msk-donskaya-omabakumova-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-2(config)#interface f0/1
msk-donskaya-omabakumova-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-omabakumova-sw-2(config-if)#^Z
msk-donskaya-omabakumova-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-2(config)#interface f0/2
msk-donskaya-omabakumova-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-omabakumova-sw-2(config-if)#
```

Рис. 15: Настройка на msk-donskaya-omabakumova-sw-2

```
msk-donskaya-omabakumova-sw-3>en
Password:
msk-donskaya-omabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-3(config)#interface fa/1
msk-donskaya-omabakumova-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

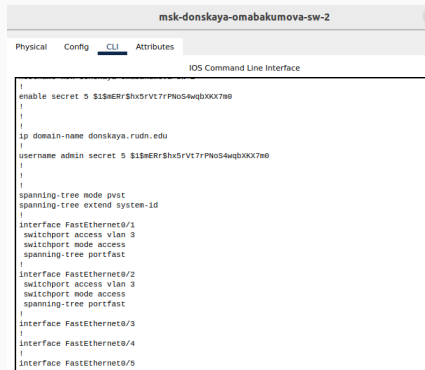
%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-omabakumova-sw-3(config-if)#^Z
msk-donskaya-omabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-sw-3(config)#interface fa/2
msk-donskaya-omabakumova-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-omabakumova-sw-3(config-if)#^Z
msk-donskaya-omabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-omabakumova-sw-3#
```

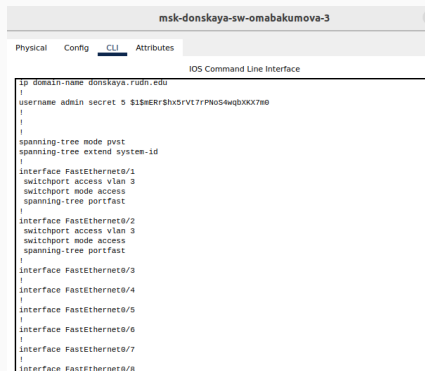
Рис. 16: Настройка на msk-donskaya-omabakumova-sw-3



The screenshot shows a web-based configuration interface for a switch named 'msk-donskaya-omabakumova-sw-2'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The configuration text is as follows:

```
enable secret 5 $1$mERr$hX5rVt7rPNoS4wqbXXX7m0
!
!
!
ip domain-name donskeya.rudn.edu
!
username admin secret 5 $1$mERr$hX5rVt7rPNoS4wqbXXX7m0
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 3
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/2
 switchport access vlan 3
 switchport mode access
 spanning-tree portfast
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
.
```

Рис. 17: Настройка прошла успешно на msk-donskaya-omabakumova-sw-2



```
msk-donskaya-sw-omabakumova-3
Physical Config CLI Attributes
IOS Command Line Interface
ip domain-name donskeya.rudn.edu
!
username admin secret 5 $1$mERR$hx5rVt7rPN0S4wqbXXX7m0
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 3
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/2
switchport access vlan 3
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
```

Рис. 18: Настройка прошла успешно на msk-donskaya-omabakumova-sw-3

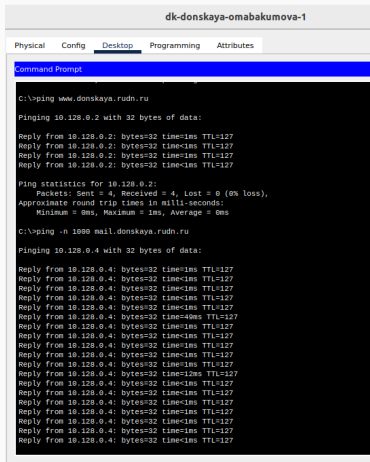


Рис. 19: Запускаем 1000 пингов

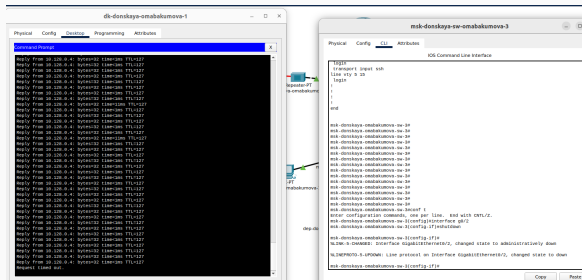


Рис. 20: Разрыв соединения, поиск резервного маршрута

Отказоустойчивость

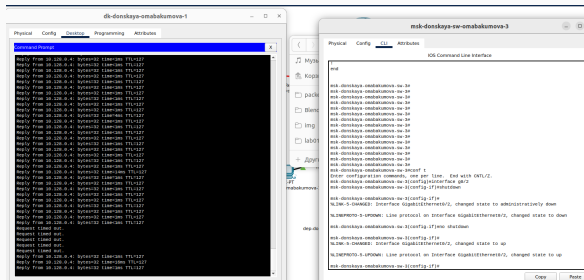


Рис. 21: Восстановление соединения, возврат на прежний маршрут

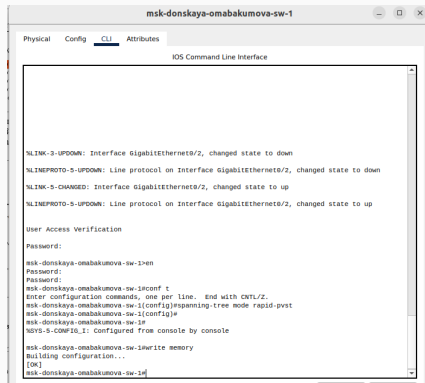


Рис. 22: Переключение msk-donskaya-omabakumova-sw-1

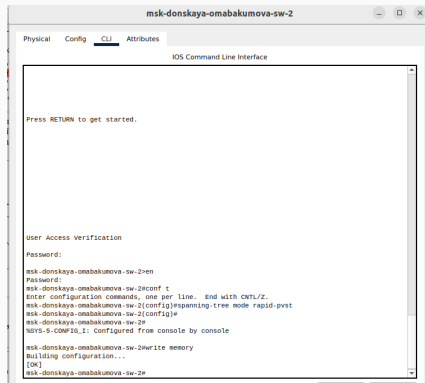


Рис. 23: Переключение msk-donskaya-omabakumova-sw-2

```
[OK]
msk-donskaya-onabakumova-sw-3#
msk-donskaya-onabakumova-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-sw-3(config)#spanning-tree mode rapid-pvst
msk-donskaya-onabakumova-sw-3(config)#
msk-donskaya-onabakumova-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-onabakumova-sw-3#
```

Рис. 24: Переключение msk-donskaya-onabakumova-sw-3

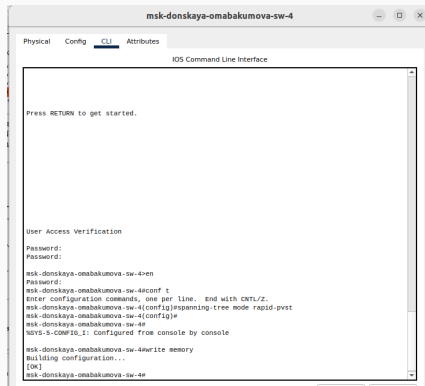
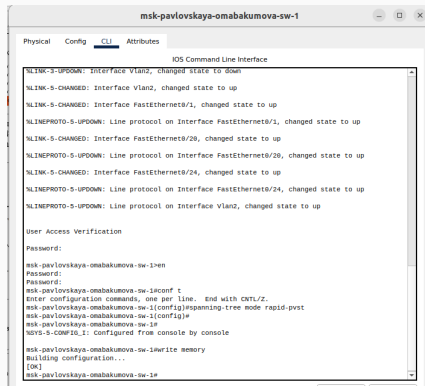


Рис. 25: Переключение msk-donskaya-omabakumova-sw-4



```
msk-pavlovskaya-omabakumova-sw-1
Physical Config CLI Attributes
IOS Command Line Interface
%LINK-3-UPDOWN: Interface Vlan2, changed state to down
%LINK-5-CHANGED: Interface Vlan2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/20, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

User Access Verification
Password:
msk-pavlovskaya-omabakumova-sw-1>en
Password:
msk-pavlovskaya-omabakumova-sw-1>conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-pavlovskaya-omabakumova-sw-1(config)#spanning-tree mode rapid-pvst
msk-pavlovskaya-omabakumova-sw-1(config)#
msk-pavlovskaya-omabakumova-sw-1#
%SYS-5-CONF:6_1: Configured from console by console

msk-pavlovskaya-omabakumova-sw-1>write memory
Building configuration...
[OK]
msk-pavlovskaya-omabakumova-sw-1#
```

Рис. 26: Переключение msk-pavlovskaya-omabakumova-sw-1

Отказоустойчивость

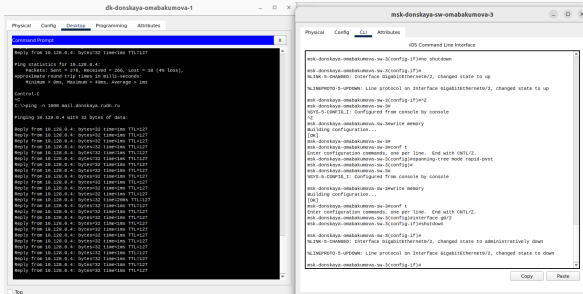


Рис. 27: Разрыв соединения, прошло незаметно

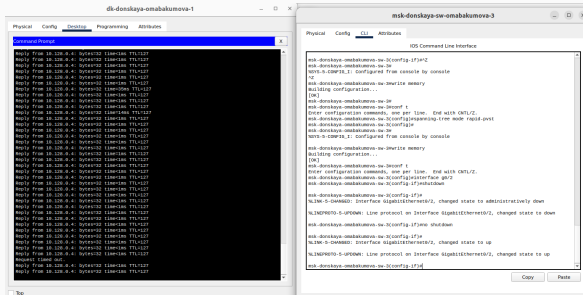


Рис. 28: Восстановление соединения, потеря буквально в один пинг

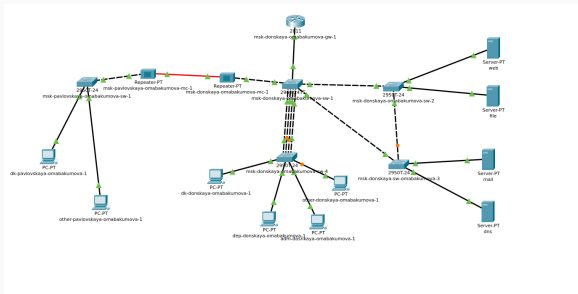
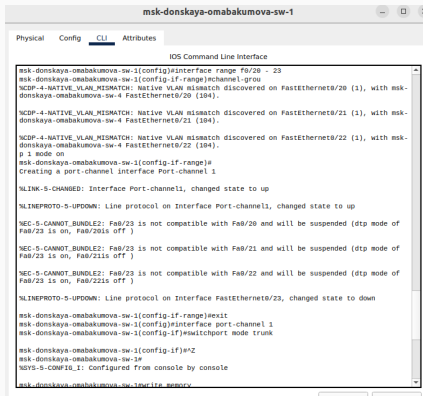


Рис. 29: Логическая схема локальной сети с агрегированным соединением



```
msk-donskaya-omabakumova-sw-1
Physical Config CLI Attributes
IOS Command Line Interface
msk-donskaya-omabakumova-sw-1(config)#interface range fa0/20 - 23
msk-donskaya-omabakumova-sw-1(config-if-range)#channel-group
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/20 (1), with msk-
donskaya-omabakumova-sw-4 FastEthernet0/20 (104).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (1), with msk-
donskaya-omabakumova-sw-4 FastEthernet0/21 (104).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/22 (1), with msk-
donskaya-omabakumova-sw-4 FastEthernet0/22 (104).
p 1 mode on
msk-donskaya-omabakumova-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/20 is off)
%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/21 is off)
%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/22 is off)
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
msk-donskaya-omabakumova-sw-1(config-if-range)#exit
msk-donskaya-omabakumova-sw-1(config)#interface port-channel 1
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk
msk-donskaya-omabakumova-sw-1(config-if)#^Z
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-omabakumova-sw-1#write memory
```

Рис. 30: Настройка агрегирования на msk-donskaya-omabakumova-sw-1

```
msk-donskaya-onabakumova-sw-4>en
Password:
msk-donskaya-onabakumova-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-sw-4(config)#int range fa/20 - 23
msk-donskaya-onabakumova-sw-4(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/20 (104), with
msk-donskaya-onabakumova-sw-1 FastEthernet0/20 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (104), with
msk-donskaya-onabakumova-sw-1 FastEthernet0/21 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/22 (104), with
msk-donskaya-onabakumova-sw-1 FastEthernet0/22 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (104), with
msk-donskaya-onabakumova-sw-1 Port-channel1 (1).

msk-donskaya-onabakumova-sw-4(config-if-range)#no switchport access vlan 104
msk-donskaya-onabakumova-sw-4(config-if-range)#exit
msk-donskaya-onabakumova-sw-4(config)#interface range fa/20 - 23
msk-donskaya-onabakumova-sw-4(config-if-range)#channel-group 1 mode on
msk-donskaya-onabakumova-sw-4(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/20is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/21is off )

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp mode of
Fa0/23 is on, Fa0/22is off )
```

Рис. 31: Настройка агрегирования на msk-donskaya-onabakumova-sw-4

Выводы

Во время выполнения данной лабораторной работы я изучила протокол STP и его модификации по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.